

How to Launch the Job Portal System

Complete operational manual to build, configure, deploy, and run both Frontend and Backend components on Apache Tomcat 11 and MySQL 8.



Prerequisites & Environment Requirements

Component	Required Version	Verification Command / Purpose
Java Development Kit	JDK 17 or JDK 21+ (LTS)	<code>java -version</code> & <code>javac -version</code>
Build Tool	Apache Maven 3.8+	<code>mvn -version</code> (Builds & packages WAR file)
Web Server / Container	Apache Tomcat 11.0.x	Jakarta EE 10 compliant container (Default Port 8080)
Database Server	MySQL Community Server 8.0+	<code>mysql --version</code> (Port 3306, Service: <code>MySQL80</code>)
Web Browser	Chrome, Edge, Firefox	HTML5 & Modern CSS3 responsive client

3-Minute Fast Track Launch Sequence

```
# 1. Start MySQL Database Service (PowerShell as Admin)
Start-Service MySQL80

# 2. Build and Package Application into deployable WAR file
mvn clean package

# 3. Deploy WAR to Tomcat 11 Webapps Directory
Copy-Item target\JobPortal.war "C:\Program Files\apache-tomcat-11.0.26\webapps\" -Force

# 4. Start Apache Tomcat Server
& "C:\Program Files\apache-tomcat-11.0.26\bin\startup.bat"

# 5. Open Web Browser and Access Website
http://localhost:8080/JobPortal
```

Frontend Architecture

CLIENT / UI

- **JSP Views:** Dynamic templates in `src/main/webapp/` rendering server data with zero business logic leakage.
- **Design System:** Curated tokens in `css/design-system.css` providing responsive cards, modals, and badge styles.
- **Client Feedback:** Client-side validation providing immediate UX feedback before form submission.

Backend Architecture

SERVER / DATA

- **Jakarta Servlets:** 17+ HTTP controllers in `controller/` handling routing, sessions, and business transactions.
- **DAO Layer:** Pure JDBC data access objects using parameterized SQL PreparedStatements to eliminate SQL injection.
- **Security:** PBKDF2 password hashing & role authorization filters.

Backend Setup: MySQL Database & Maven Build

Follow these precise steps to configure MySQL persistence and build the deployable WAR artifact.

1 Initialize the MySQL Database & Schema

Ensure the MySQL 8.0 server service is running, create the database `job_portal_db`, and execute the SQL schema definition:

```
# Step 1A: Start MySQL Service in PowerShell (if not running)
Start-Service MySQL80

# Step 1B: Log into MySQL CLI
mysql -u root -p

# Step 1C: Inside MySQL prompt: Create Database and Run Schema
CREATE DATABASE IF NOT EXISTS job_portal_db;
USE job_portal_db;
SOURCE c:/Learning some new stuf/Java Mini Project/database/schema.sql;
SHOW TABLES;
```

i Created Tables: `users` (login & PBKDF2 password hashes), `jobseeker_profile`, `recruiter_profile`, `jobs`, and `applications`. Foreign keys maintain relational integrity via `ON DELETE CASCADE`.

2 Configure Connection Properties (`db.properties`)

The backend reads database connection parameters from `src/main/resources/db.properties`. Verify that the file contains your active MySQL credentials:

```
# File: src/main/resources/db.properties
db.url=jdbc:mysql://localhost:3306/job_portal_db?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC
db.user=root
db.password=Harry@12@
db.driver=com.mysql.cj.jdbc.Driver
```

⚠ Credential Synchronization: If your local MySQL root password differs, update `db.password` or provide it at runtime using `$env:DB_PASSWORD="YourPassword"` before packaging.

3 Compile & Package with Maven

Execute Maven to compile all Java source files, bundle JSP/CSS frontend resources, and package the single `JobPortal.war` archive:

```
# In the project workspace root:
mvn clean package
```

```
[INFO] --- compiler:3.13.0:compile (default-compile) @ JobPortal ---
[INFO] Compiling 32 source files with javac [debug release 17] to target/classes
[INFO] --- war:3.4.0:war (default-war) @ JobPortal ---
[INFO] Building war: C:\Learning some new stuf\Java Mini Project\target\JobPortal.war
[INFO] BUILD SUCCESS
```

Deploying to Apache Tomcat 11

Deploy the compiled WAR package into Apache Tomcat 11 and start the web application server.

4 Deploy `JobPortal.war` to Tomcat `webapps/`

Copy the built WAR from `target/` to the Tomcat `webapps/` directory. Tomcat will extract the archive and serve it under context path `/JobPortal`.

```
# PowerShell Deployment Command:
Copy-Item "target\JobPortal.war" "C:\Program Files\apache-tomcat-11.0.26\webapps\JobPortal.war" -Force

# Command Prompt (cmd.exe) Alternative:
copy /Y target\JobPortal.war "C:\Program Files\apache-tomcat-11.0.26\webapps\"
```

5 Start the Tomcat Web Server

Launch Tomcat using the startup script. A console window will launch displaying server initialization logs:

```
# Start Tomcat in a background window:
& "C:\Program Files\apache-tomcat-11.0.26\bin\startup.bat"

# Or run directly in foreground for debugging:
& "C:\Program Files\apache-tomcat-11.0.26\bin\catalina.bat" run
```

✓ **Verification Log:** Look for: `[main] org.apache.catalina.startup.Catalina.start Server startup in [XXXX] milliseconds`. This confirms the server is ready to accept HTTP traffic.

Application URL & Endpoint Mappings

Module	Public URL Path	Backend Controller / View	Access Level
Landing Page	<code>http://localhost:8080/JobPortal/</code>	<code>index.jsp</code>	PUBLIC
Registration	<code>/JobPortal/register</code>	<code>controller.RegisterServlet</code>	PUBLIC
Authentication	<code>/JobPortal/login</code>	<code>controller.LoginServlet</code>	PUBLIC
Job Seeker Area	<code>/JobPortal/jobseeker/profile</code>	<code>controller.ProfileServlet</code>	JOB SEEKER
Job Search & Apply	<code>/JobPortal/jobseeker/search-jobs</code>	<code>controller.JobSearchServlet</code>	JOB SEEKER
Recruiter Dashboard	<code>/JobPortal/recruiter/manage-jobs</code>	<code>controller.ManageJobsServlet</code>	RECRUITER
Applicant Review	<code>/JobPortal/recruiter/applicants</code>	<code>controller.RecruiterApplicantsServlet</code>	RECRUITER
Session Logout	<code>/JobPortal/logout</code>	<code>controller.LogoutServlet</code>	AUTHENTICATED

How to Gracefully Stop or Restart Tomcat

SHUTDOWN

```
# Gracefully stop Tomcat server:
& "C:\Program Files\apache-tomcat-11.0.26\bin\shutdown.bat"

# If Tomcat process remains locked on port 8080:
Get-Process -Name java | Stop-Process -Force
```

Frontend Navigation & Role Workflows

Open your web browser at <http://localhost:8080/JobPortal> to interact with the system.

Job Seeker Experience

CANDIDATE

Job Portal

Alex MorganJOB SEEKERDashboardProfileSearch JobsMy ApplicationsLogout

Explore Job Opportunities

Find relevant positions matching your technical skills, desired title, and preferred location.

KEYWORD

SKILLS

LOCATION

Java

e.g. Java, SQL, React

City, State, or Remote

Search

Clear

3 Jobs found (Filtered results)

Keyword: Java

Senior Java Backend Engineer

San Francisco, CA (Hybrid) • Posted 19 Sept 2026

REQUIRED SKILLS

JOB ID #101

Skill & Keyword Job Search UI

Step-by-Step Flow:

1. Register:

Choose *Job Seeker* role on `/register`.

2. Login:

Enter credentials on `/login`.

3. Search Jobs:

Filter by keyword, skills, or location.

4. Apply:

Submit application with a single click.

5. Track Status:

Monitor real-time status in *My Applications* (`APPLIED`, `UNDER REVIEW`, `SHORTLISTED`, `REJECTED`).

Recruiter Experience

HIRING MANAGER

Job Portal

Apex Global TechnologiesRECRUITERDashboardProfilePost JobManage JobsApplicantsLogout

Applicant Dashboard

Review candidates who have applied for your published career opportunities.

Filter by Job Vacancy:

#101: Senior Java Backend Engineer (San Francisco)

Filter Applications

Clear Filter

Applications for: Senior Java Backend Engineer (Job #101)

3 CANDIDATES

Alex Morgan

Applied for: Senior Java Backend Engineer (San Francisco, CA (Hybrid))

UNDER REVIEW

EMAIL ADDRESS

CONTACT PHONE

CANDIDATE LOCATION

EDUCATION

Dedicated Applicant Status Tracking Dashboard

Step-by-Step Flow:

1. Register:

Choose *Recruiter* role & enter company details.

2. Post Job:

Create vacancies with title, salary, skills, & type.

3. Manage Jobs:

Edit or delete active listings in real time.

4. Review Candidates:

Access the Applicant Dashboard to inspect profiles, education, and skills.

5. Update Status:

Transition candidates across the recruitment lifecycle.

Live Interface Previews

Job Portal

RegisterLogin

Welcome to Job Portal

Connecting job seekers with leading Recruiters.

Get Started — Register Now

Account Login

Second Year Academic Mini Project (P57)

Seniors • 10P • 1200 • 10/10/22

1. Landing Page (index.jsp)

Job Portal

HomeLogin

Create an Account

Join the Job Portal to become a user or hire talented talent.

1. Select Account Type

Job Seeker

Recruiter

2. Account Information

Full Name *

David Address *

Phone *

2. Atomic Registration

Job Portal

Apex Global TechnologiesRECRUITERDashboardProfilePost JobManage JobsApplicantsLogout

Manage Job Postings

Review, update, and manage the career vacancies you have published.

Showing 3 active job postings

Senior Java Backend Engineer

San Francisco, CA (Hybrid) • Posted Sept 19, 2026 • Job #101

View Details

Details

Post

Cancel

Reject

Shortlist

Mark as Favorite

We are seeking an experienced Java Backend Engineer to develop robust, scalable enterprise services, high-throughput REST APIs, and database integrations. Responsibilities include building transactional database layer, supervising query execution, and collaborating with cross-functional product team.

Oversee New Developer (1 FTE)

Post Vacancies

Full Job

Temporary

Full Stack Web Developer

ACTIVE VACANCY

3. Recruiter Job Management

Job Portal System • Launch & Operations Guide

Page 4 of 5

Troubleshooting & Diagnostic Cheatsheet

Solutions for common environment, database, port, and deployment issues.

Issue 1: "Registration failed due to a database error" (SQLException 1045)

DATABASE AUTH

Cause: Password mismatch in `src/main/resources/db.properties` or MySQL service is stopped.

Solution: Verify MySQL service status and synchronize credentials:

```
Start-Service MySQL80
# Ensure db.properties contains valid credentials, then rebuild & redeploy:
mvn clean package
Copy-Item target\JobPortal.war "C:\Program Files\apache-tomcat-11.0.26\webapps\" -Force
```

Issue 2: Port 8080 Conflict (Tomcat Fails to Start)

PORT CONFLICT

Cause: Another service or zombie Java process is holding port 8080.

Solution: Identify and terminate the occupying process, or adjust port in `conf/server.xml`:

```
# Find process occupying port 8080:
netstat -ano | findstr :8080
# Terminate process by PID (e.g., PID 1234):
taskkill /F /PID 1234
```

Issue 3: HTTP 404 Not Found on `/JobPortal`

CONTEXT ERROR

Cause: The WAR file was not placed in `webapps/` or Tomcat failed during WAR extraction.

Solution: Check `logs/catalina.out` and confirm the directory `webapps/JobPortal` exists with extracted JSPs and WEB-INF classes.

Full Launch Checklist

✓ Database Service	MySQL 8.0 running on <code>localhost:3306</code> with <code>job_portal_db</code> initialized.
✓ Configuration	<code>src/main/resources/db.properties</code> contains valid database username and password.
✓ Compilation	Maven build ran cleanly with <code>BUILD SUCCESS</code> producing <code>JobPortal.war</code> .
✓ Deployment	<code>JobPortal.war</code> copied to Tomcat 11 <code>webapps/</code> directory.
✓ Server Running	Tomcat 11 active on <code>localhost:8080</code> without port collision.
✓ Website Live	Browser successfully loads <code>http://localhost:8080/JobPortal</code> with all role flows active.